

SUSTAINABLE SUPPLY CHAIN FOR DRIED MANGO FROM AFRICA TO GERMANY

WEEK 4

FARM SECURITY ZONING

From Risk & Asset Analysis to a Clear Spatial Security Structure

Fencing & Security App

Prepared by Tania | Excellence Program — Week 4

A preliminary working model for discussion and refinement

1 Why farm security zoning?

Week 3 identified the people, assets, activities and operating conditions that can affect farm security. The next step is to translate that understanding into a clearer spatial structure. A farm should not be treated as one uniform security area, because activities and assets do not carry the same level of sensitivity.

FROM WEEK 3 TO WEEK 4

Risk & Assets

What needs protection?



Security Needs

How sensitive is it?



Farm Zoning

How should movement work?

What Week 4 does

It identifies the main functional areas of the farm, gives them a preliminary security sensitivity level, and makes movement between those areas easier to discuss.

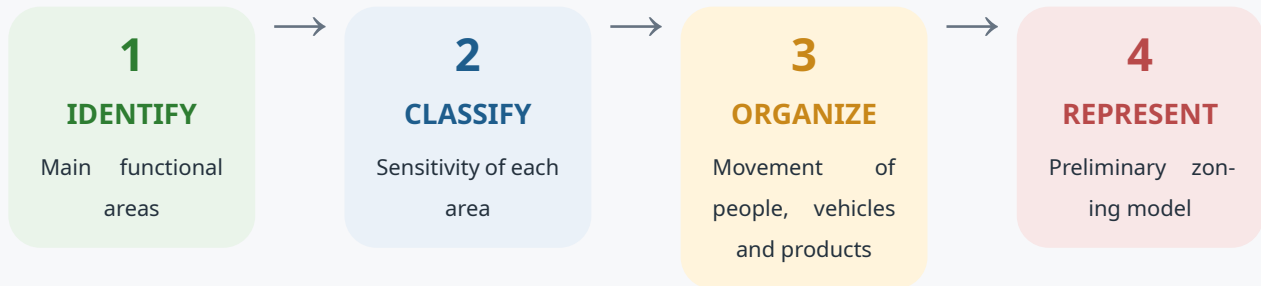
What Week 4 produces

It produces a preliminary zoning model: an illustrative working structure that can later be refined with the real farm layout and operational information.

A practical principle Zoning must also consider the physical environment in which the security system operates, including heat, dust, weather, power reliability and connectivity limitations.

2 The four-step zoning method

The zoning model is not arbitrary. It follows a simple method that moves from understanding the farm to representing a first spatial security structure. Each step answers a clear question and creates an observable result for discussion.



1. Identify

List the farm areas according to what happens there: entry and reception, production, operations, storage, equipment, loading and critical infrastructure.

2. Classify

Consider the sensitivity of each area. The question is not only what is located there, but also what impact unauthorized entry or disruption could have.

3. Organize movement

Consider how workers, visitors, vehicles and products should move. Movement should become more controlled as sensitivity increases.

4. Represent

Use a clear preliminary model to make the security logic visible, reviewable and easier to refine later.

IDENTIFY → CLASSIFY → ORGANIZE MOVEMENT → CREATE ZONING MODEL

3 Four preliminary security levels

The following model uses four simple levels. It is a preliminary classification for discussion, not a final farm layout or an access-control specification. The levels help distinguish between areas where movement can be more frequent and areas where access should be more strictly limited.

ZONE 1 — OPEN

Examples: entrance, reception and designated visitor area.

ZONE 2 — CONTROLLED

Examples: production areas and staff operational areas.

ZONE 3 — RESTRICTED

Examples: warehouse, storage, equipment and loading-related sensitive areas.

Purpose: protect more valuable assets and sensitive

ZONE 4 — CRITICAL

Examples: critical infrastructure and assets whose compromise could have a major operational impact.

Purpose: strictly controlled access.

Sensitivity increases across the model



The model does not assume that every farm has the same physical arrangement. It establishes a shared language for discussing the protection needs of different activities, assets and operational areas.

4 Movement and access

Zoning is not simply drawing colored boxes on a map. It must guide how people, vehicles and products move through the farm. The principle is that access becomes more controlled as the sensitivity of the destination area increases.

Workers

Workers need regular access to operational areas. Their movement should support daily work while remaining connected to the purpose of each zone.

Visitors

Visitors should remain concentrated around designated entrance and reception areas as much as possible, rather than moving through sensitive operations.

Vehicles & products

Loading and dispatch should be organized so that vehicles and products do not create unnecessary access to more sensitive areas.

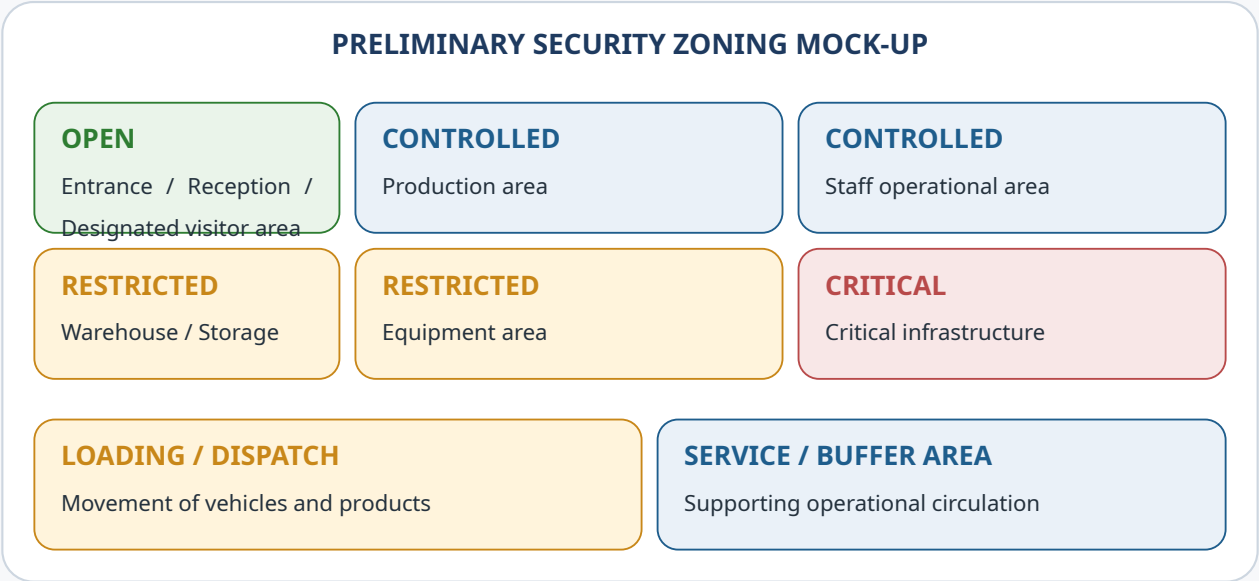
Conceptual movement principle

ENTRANCE ↓ OPEN ↓ CONTROLLED ↓ RESTRICTED ↓ CRITICAL

A conceptual progression, not a claim about the final physical arrangement of the farm.

Design implication Visitors, workers, vehicles and products do not require the same paths or the same access to every part of the farm. Zoning makes this difference visible before later physical-security decisions are developed.

5 Preliminary Security Zoning Mock-up



Colour key

- OPEN ■ CONTROLLED
- RESTRICTED ■ CRITICAL

Illustrative working model

To be refined using the real farm layout and operational information.

This mock-up makes the proposed security logic observable. It provides a starting point for reviewing whether the future farm layout, activities and routes support an appropriate separation between open, operational, sensitive and critical areas.

6 Week 4 result

Week 4 establishes the spatial security logic that later physical-security decisions can build upon. It does not yet create a final fencing design, surveillance design, access-control specification or farm architecture. Instead, it makes the relationship between areas, sensitivity and movement visible.

Established

The farm is understood as a set of functional areas rather than one uniform security space.

Connected

The zoning work continues from the Week 3 risk and asset analysis and gives it a spatial form.

Organized

Workers, visitors, vehicles and products are considered as distinct movement categories.

Prepared

The preliminary model creates a clearer basis for later physical-security design and review.

KEY TAKEAWAY

The farm should not be treated as one uniform security area.

Different activities, assets and risks require different levels of protection and controlled movement.

Continuity note: This Week 4 submission follows the review and requirements work completed in Weeks 1–2 and the risk and asset analysis completed in Week 3. Its purpose is to produce a clear, preliminary zoning model that can be discussed and refined with real farm information.